

LESSBURG, VA, USA

WMO: 724055

Lat: 39.078N

Lon: 77.558W

Elev: 389

StdP: 14.49

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 03714

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 13.9	(c) 18.1	(d) -3.3	(e) 4.7	(f) 22.0	(g) 1.4	(h) 6.0	(i) 22.3	(j) 28.2	(k) 32.9	(l) 25.4	(m) 32.7	(n) 7.1	(o) 320	(p) 0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.7	(c) 94.9	(d) 76.6	(e) 91.4	(f) 75.1	(g) 90.2	(h) 74.5	(i) 79.4	(j) 90.1	(k) 78.0	(l) 88.1	(m) 76.8	(n) 86.2	(o) 6.7	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.9	142.1	84.3	75.1	133.5	82.5	73.3	125.6	80.8	43.1	89.5	41.7	87.1	40.4	86.8	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.2	(b) 19.1	(c) 16.5	(e) 7.2	(f) 98.8	(g) 5.2	(h) 2.1	(i) 3.4	(j) 100.3	(k) 0.4	(l) 101.5	(m) -2.5	(n) 102.7	(o) -6.3	(p) 104.2		
(4) 23.2	(5) 19.1	(6) 16.5	(7) 7.2	(8) 98.8	(9) 5.2	(10) 2.1	(11) 3.4	(12) 100.3	(13) 0.4	(14) 101.5	(15) -2.5	(16) 102.7	(17) -6.3	(18) 104.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.7	34.7	37.2	45.3	56.0	65.1	73.9	78.0	76.7	69.5	57.8	46.5	38.5		
	DBStd	17.25	10.01	9.53	9.46	8.83	7.76	6.15	4.99	5.04	6.76	8.07	8.46	8.51		
	HDD50	1658	486	370	208	37	1	0	0	0	0	18	165	373		
	HDD65	4396	938	777	613	296	98	9	0	0	33	253	556	823		
	CDD50	4107	13	13	63	216	470	716	869	828	585	259	60	15		
	CDD65	1369	0	0	3	25	102	275	404	363	168	28	1	0		
	CDH74	13621	0	6	49	361	1059	2678	4219	3546	1416	271	14	2		
	CDH80	5623	0	0	9	119	373	1091	1931	1535	498	65	2	0		
	WSAvg	5.6	6.8	7.0	7.3	6.8	5.2	4.7	4.2	3.9	4.4	5.0	5.8	5.9		
	PrecAvg	40.7	2.8	2.3	3.4	3.5	4.0	3.7	3.8	3.4	4.0	3.3	2.9	3.2		
(15) Precipitation	PrecMax	61.3	6.9	5.7	7.3	7.9	8.8	8.3	8.1	6.3	9.4	7.3	6.8	5.6		
	PrecMin	28.7	1.3	0.6	0.0	1.7	1.1	1.3	1.6	1.5	0.0	0.1	0.6	0.9		
	PrecStd	7.4	1.4	1.2	1.5	1.5	1.9	1.9	1.5	1.1	2.5	2.0	1.4	1.3		
	PrecStd	7.4	1.4	1.2	1.5	1.5	1.9	1.9	1.5	1.1	2.5	2.0	1.4	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.7	72.2	80.7	90.3	91.3	96.6	99.0	97.4	93.2	86.2	75.1	70.0		
		MCWB	60.4	59.8	62.9	67.5	73.2	77.3	77.2	77.5	73.7	72.1	62.3	61.9		
	2%	DB	62.7	63.6	72.7	82.4	88.3	91.5	95.2	93.4	89.9	81.3	70.0	62.6		
		MCWB	57.3	52.7	58.3	64.6	72.1	75.5	77.5	76.3	73.1	68.1	60.5	57.1		
	5%	DB	55.2	56.9	67.5	77.5	84.0	90.0	91.4	90.8	86.0	76.6	64.3	55.5		
		MCWB	49.9	48.0	55.8	62.6	69.9	74.7	75.8	75.1	71.8	66.1	57.3	49.6		
	10%	DB	48.4	52.0	62.8	72.4	80.7	86.4	90.1	88.3	81.8	72.3	61.2	52.0		
		MCWB	44.2	44.9	52.6	59.8	67.7	73.2	75.3	74.0	69.7	63.6	54.2	46.7		
	0.4%	WB	63.1	62.5	64.6	70.5	76.3	80.1	81.4	81.2	78.4	74.0	66.5	64.0		
		MCDB	64.0	69.7	75.5	81.5	88.1	91.8	93.5	92.1	87.6	80.2	72.1	66.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	57.8	55.5	61.5	67.6	74.1	77.9	79.6	79.1	76.3	71.7	63.3	58.7		
		MCDB	62.3	61.3	70.0	78.9	84.6	88.7	90.6	89.4	84.3	78.6	66.5	62.5		
	5%	WB	51.0	49.8	57.4	64.7	72.0	76.5	78.2	77.5	74.2	68.6	59.5	52.1		
		MCDB	55.8	55.6	64.8	74.0	81.1	86.2	88.4	87.0	80.7	74.3	63.8	55.1		
	10%	WB	45.0	45.6	53.4	61.8	69.7	74.9	77.0	76.0	72.4	64.8	55.4	46.4		
		MCDB	47.8	49.4	61.1	70.5	76.9	83.4	86.4	84.7	78.6	70.1	60.1	49.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	17.6	19.9	22.2	21.0	20.3	20.7	20.6	20.4	20.6	18.8	15.7		
		MCDBR	23.4	26.4	29.2	30.5	27.1	24.5	23.2	23.8	25.4	26.5	25.3	22.5		
		MCWBR	18.4	18.4	17.2	15.3	12.7	10.1	8.9	9.0	10.6	14.5	17.0	17.5		
		MCDBR	21.6	23.5	24.9	25.3	23.9	21.4	20.7	21.0	19.9	21.2	20.3	20.2		
	5% WB	MCWBR	18.4	18.1	17.0	14.1	11.8	9.7	9.0	8.9	9.7	12.9	16.1	17.8		
		MCWBR	18.4	18.1	17.0	14.1	11.8	9.7	9.0	8.9	9.7	12.9	16.1	17.8		
(40) Clear-Sky Solar Irradiance	taub		0.305	0.321	0.356	0.400	0.440	0.504	0.518	0.518	0.432	0.362	0.322	0.307		
	taud		2.464	2.397	2.372	2.272	2.191	2.049	2.029	2.010	2.229	2.425	2.499	2.492		
	Ebn at Noon		276	286	285	278	268	250	245	241	257	267	268	267		
	Edh at Noon		25	31	35	42	46	53	54	53	40	30	24	23		
	RadAvg		639	911	1210	1533	1722	1902	1866	1667	1363	1017	733	561		
(45) All-Sky Solar Radiation	RadStd		56	104	77	119	181	144	104	84	170	115	62	56		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	+0.68	+1.24	N/A	N/A	N/A	+139
Regional (65 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+225	+124

Nomenclature: See separate page